29

LLMs

Large Language Models — the brains behind modern chatbots

What they really are, and how to use them well

AI/ML Engineer Learning Series · Deep Dive Edition

What it is	A huge Transformer trained on massive text
Full name	Large Language Model
Examples	GPT, Claude, Llama, Gemini
Core skill	Predicting the next word, extremely well
You use them	Groq powers your RAG chatbot's answers

What's Inside

Here is everything covered in this guide, in order:

1. What is an LLM?

An LLM, or Large Language Model, is a very big Transformer (Topic 20) trained on an enormous amount of text — much of the public internet, books, and code. 'Large' means it has billions of weights and was trained on trillions of words. The result is a model that can read and write human language remarkably well.

LLMs are the brains behind chatbots like ChatGPT, Claude, and Gemini. They can answer questions, write essays and code, summarise, translate, and hold conversations. They feel intelligent — but underneath, they're doing something surprisingly simple, which we'll see next.

■ *You already use LLMs: your RAG chatbot uses Groq (a fast LLM provider) to write its final answers. This topic explains what's happening inside that LLM and how to get the best from it.*

2. Just Predicting the Next Word

Here's the secret that surprises everyone: at its core, an LLM just predicts the next word. You give it some text, and it predicts what word most likely comes next. Then it adds that word and predicts the next. Over and over. That's it.

An LLM just predicts the next word, over and over

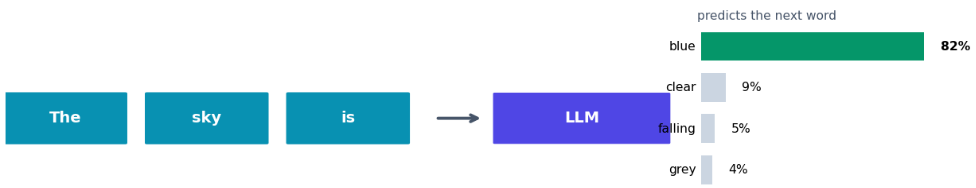

Fig 1 — Given 'The sky is', the model predicts 'blue' as most likely, adds it, then predicts again.

Give it 'The sky is' and it predicts 'blue' with high probability. It writes essays, code, and answers the same way — one word at a time, each chosen as the most fitting next word. How can something so simple seem so smart? Because to predict the next word really well across all of human text, the model had to learn grammar, facts, reasoning patterns, and style. Good prediction requires real understanding of language.

3. The Power of Scale

Why did LLMs suddenly get so good around 2020? One word: scale. Researchers found that making models bigger — more weights, more data, more compute — kept making them dramatically smarter, in ways nobody fully expected. Abilities like reasoning and coding 'emerged' as the models grew.

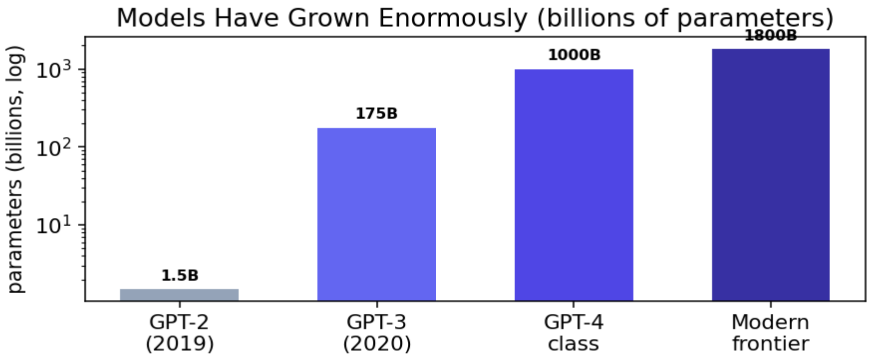

Fig 2 — Models grew from millions to over a trillion parameters in just a few years (note the log scale).

GPT-2 in 2019 had 1.5 billion parameters. GPT-3 jumped to 175 billion. Frontier models now have far more. Each big jump in scale unlocked new abilities. This 'scaling' insight — that bigger reliably means better — drove the whole LLM boom. It also means these models are extremely expensive to train, which is why most people use ones built by big labs.

4. How a Chatbot is Built

A raw LLM that just predicts text isn't yet a helpful assistant. Turning it into a chatbot like Claude takes three stages:

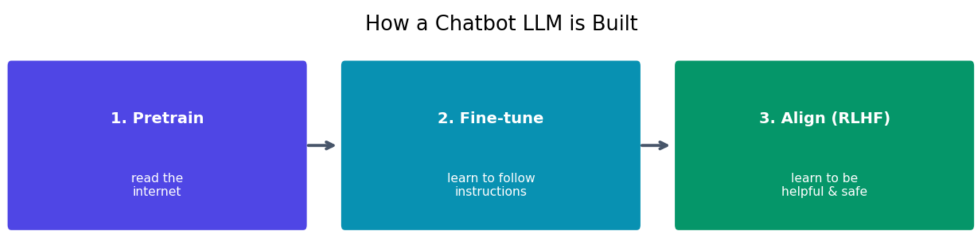

Fig 3 — Three stages: pretrain on text, fine-tune to follow instructions, align to be helpful and safe.

Stage	What happens
1. Pretraining	Read huge amounts of text — learns language and facts
2. Instruction fine-tuning	Learn to follow instructions and answer helpfully
3. Alignment (RLHF)	Learn from human feedback to be helpful, honest, safe

RLHF (Reinforcement Learning from Human Feedback) is the famous third stage: humans rate the model's answers, and the model learns to give the kinds of answers people prefer. This is what turns a raw text-predictor into a polite, helpful assistant.

5. The Context Window

The context window is how much text an LLM can 'see' at once — its working memory. It includes your prompt plus everything in the current conversation. Measured in tokens (remember: about 3/4 of a word each).

If a conversation gets longer than the context window, the model starts 'forgetting' the earliest parts — they fall out of its view. This is why a chatbot can lose track of something said much earlier in a very long chat. Modern models have large windows (hundreds of thousands of tokens), but the limit always exists.

Context window	Roughly holds
8,000 tokens	A long article
128,000 tokens	A whole book
1,000,000 tokens	Several books at once

■ *The context window is why RAG exists. Instead of stuffing a whole PDF into the prompt, your RAG chatbot finds just the few relevant chunks and puts only those in the context window. Smart use of limited space.*

6. Temperature — Controlling Creativity

When an LLM picks the next word, you can control how 'adventurous' it is with a setting called temperature. Low temperature makes it pick the most likely word almost every time — focused and predictable. High temperature lets it pick less likely words — more creative and varied.

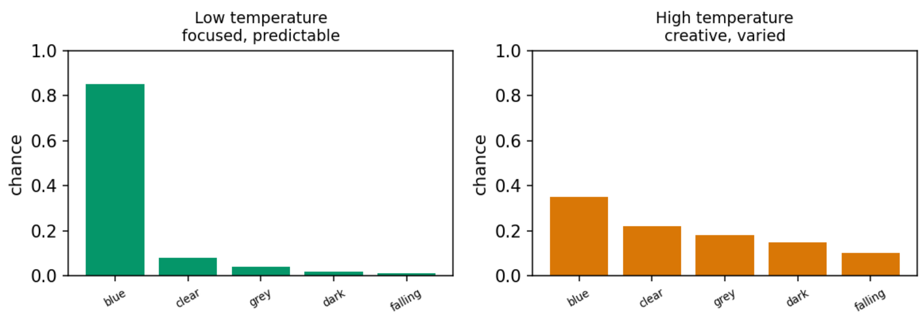

Fig 4 — Low temperature is focused and consistent. High temperature is more creative and surprising.

Temperature	Best for
Low (0 - 0.3)	Facts, code, math — you want consistency
Medium (0.5 - 0.7)	General chat, balanced answers
High (0.8 - 1.2)	Creative writing, brainstorming, variety

■ *For your RAG chatbot and anything factual, use a low temperature. You want accurate, consistent answers based on the retrieved text — not creative invention. Save high temperature for brainstorming and creative tasks.*

7. What LLMs are Great (and Bad) At

Great at	Not so good at
Writing and rewriting text	Exact math and counting
Summarising and explaining	Knowing very recent events
Coding help and debugging	Real-time or private data
Translation and language tasks	Being reliably factual on details
Brainstorming and ideas	True logical certainty
Answering general questions	Knowing what it doesn't know

LLMs are amazing language tools but not databases or calculators. They predict plausible text, which is perfect for writing and explaining, but risky for precise facts or fresh information. The fix for many of these weaknesses is to GIVE them the right information (RAG) or tools (agents) — the next stages of your learning.

8. Hallucinations

A hallucination is when an LLM confidently states something false — a made-up fact, a fake citation, a wrong date. It happens because the model predicts plausible-sounding text, and sometimes the most plausible-sounding text isn't true. It isn't lying; it genuinely doesn't know the difference.

This is the single most important limit to remember. Always verify important facts from an LLM. The best defences are: give it the real information to work from (RAG), ask it to cite sources you can check, and use a low temperature for factual tasks.

■ *RAG (next topic) is largely a hallucination fix: by giving the LLM the actual relevant documents, you ground its answers in real text instead of letting it guess. This is exactly why your PDF RAG chatbot is more trustworthy than a raw LLM.*

9. Open vs Closed Models

There are two kinds of LLMs you can use, and knowing the difference helps you choose.

	Closed (API)	Open (download)
Examples	GPT, Claude, Gemini	Llama, Mistral, Qwen
How to use	Call their API	Run it yourself
Quality	Usually top-tier	Very good, improving fast
Cost	Pay per use	Free to run (need hardware)
Privacy	Data goes to provider	Stays on your machine
Control	Limited	Full — you can fine-tune it

Closed models via API (like the Groq-hosted models in your chatbot) are easiest and often highest quality — you just call them. Open models you download and run yourself: more setup, but free, private, and fully under your control (you can even fine-tune them). Many projects use closed APIs to start, then move to open models for cost or privacy.

10. Common Mistakes

Mistake	Why it's a problem	Fix
Trusting facts blindly	LLMs hallucinate	Verify; use RAG and citations
High temp for facts	Inconsistent, invented answers	Low temperature for factual work
Overflowing context	Early text is forgotten	Keep prompts focused; use RAG
Using it as a calculator	Bad at exact math	Use real tools / code for math
Expecting recent knowledge	It has a cutoff date	Give it fresh info via RAG/tools
Vague prompts	Weak answers	Be clear and specific (next topic)

Quick Reference Cheatsheet

Concept	Meaning
LLM	huge Transformer trained on massive text
Core skill	predict the next word, repeatedly
Scale	bigger models = smarter (the big insight)
Pretraining	learn language from huge text
RLHF	human feedback makes it helpful & safe

Context window	how much text it can see at once
Token	~3/4 of a word
Temperature	low=focused, high=creative
Hallucination	confidently stating something false
RAG	give it real docs to reduce hallucination
Closed model	API (GPT, Claude) — easy, top quality
Open model	download (Llama) — free, private, tunable

The one idea to remember: an LLM is a giant Transformer that predicts the next word incredibly well. That simple skill, scaled up massively, produces something that can write, code, and converse. Just remember it predicts plausible text — so verify facts and ground it with RAG.

Next Topic: Prompt Engineering — the practical skill of writing instructions that get the best results from an LLM. Small changes in how you ask can dramatically improve the answers.